

Lab - 5

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Question 1:

Code:

```
main.cpp
1  #include <iostream>
2  #include <vector>
3  using namespace std;
4
5  void merge(vector<int>& a, int l, int m, int r) {
6      vector<int> tmp;
7      int i = l, j = m + 1;
8      while (i <= m && j <= r) tmp.push_back(a[i] < a[j] ? a[i++] : a[j++]);
9      while (i <= m) tmp.push_back(a[i++]);
10     while (j <= r) tmp.push_back(a[j++]);
11     for (int k = 0; k < tmp.size(); ++k) a[l + k] = tmp[k];
12 }
13
14 void mergeSort(vector<int>& a, int l, int r) {
15     if (l >= r) return;
16     int m = l + (r - l) / 2;
17     mergeSort(a, l, m);
18     mergeSort(a, m + 1, r);
19     merge(a, l, m, r);
20 }
21
22 int main() {
23     vector<int> a = {38, 27, 43, 3, 9, 82, 10};
24     mergeSort(a, 0, a.size() - 1);
25     for (int x : a) cout << x << " ";
26     return 0;
27 }
```

Output:

```
3 9 10 27 38 43 82
...Program finished with exit code 0
Press ENTER to exit console.
```

Question 2:

Code:

```
main.cpp
1  #include <iostream>
2  #include <vector>
3  #include <algorithm>
4  using namespace std;
5
6  void bucketSort(vector<float>& a) {
7      int n = a.size();
8      vector<vector<float>>> b(n);
9      for (int i = 0; i < n; ++i) b[int(n * a[i])].push_back(a[i]);
10     for (int i = 0; i < n; ++i) sort(b[i].begin(), b[i].end());
11     a.clear();
12     for (int i = 0; i < n; ++i)
13         for (float x : b[i]) a.push_back(x);
14 }
15
16 int main() {
17     vector<float> a = {0.78, 0.17, 0.39, 0.26, 0.72, 0.94, 0.21};
18     bucketSort(a);
19     for (float x : a) cout << x << " ";
20     return 0;
21 }
22
```

Output:

```
0.17 0.21 0.26 0.39 0.72 0.78 0.94

...Program finished with exit code 0
Press ENTER to exit console.
```

Question 3:

Code:

```
main.cpp
1  #include <iostream>
2  #include <vector>
3  using namespace std;
4
5  int partition(vector<int>& a, int l, int h) {
6      int pivot = a[h], i = l - 1;
7      for (int j = l; j < h; ++j)
8          if (a[j] < pivot) swap(a[++i], a[j]);
9      swap(a[i + 1], a[h]);
10     return i + 1;
11 }
12
13 void quickSort(vector<int>& a, int l, int h) {
14     if (l < h) {
15         int p = partition(a, l, h);
16         quickSort(a, l, p - 1);
17         quickSort(a, p + 1, h);
18     }
19 }
20
21 int main() {
22     vector<int> a = {10, 7, 8, 9, 1, 5};
23     quickSort(a, 0, a.size() - 1);
24     for (int x : a) cout << x << " ";
25     return 0;
26 }
27
```

Output:

```
1 5 7 8 9 10
...Program finished with exit code 0
Press ENTER to exit console.
```